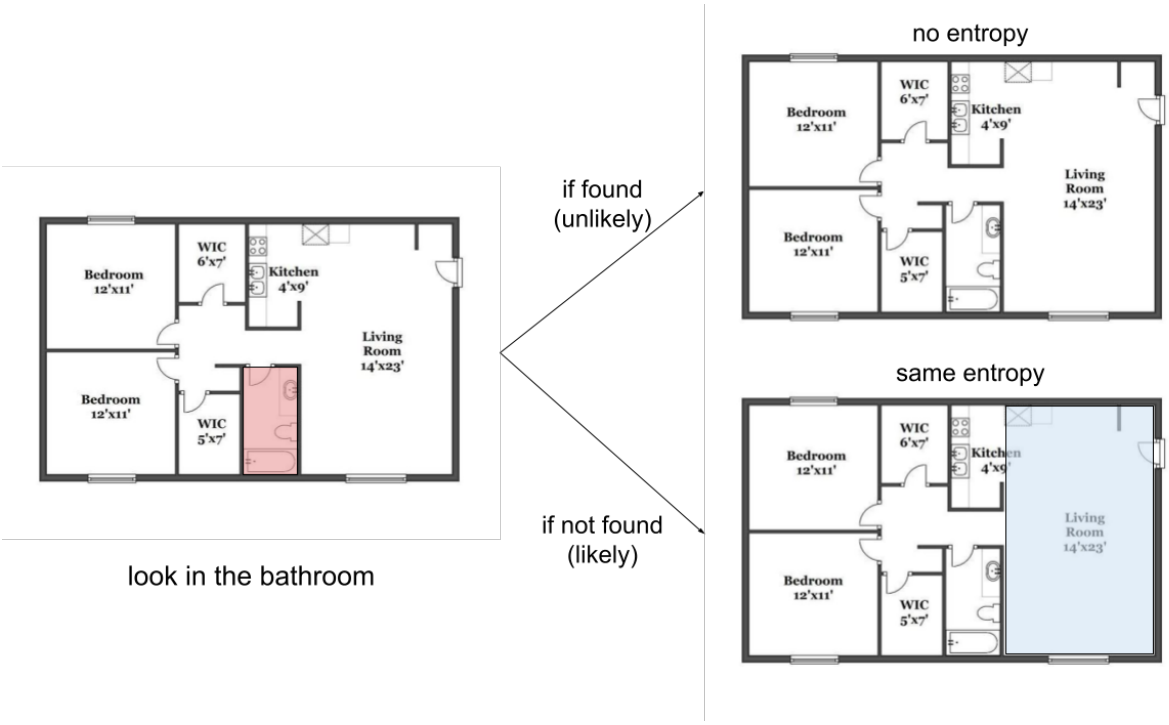


Figure 6.4 Entropy of the location of the remote after the bathroom is searched. As the remote cannot be found in the bathroom, the entropy in the posterior distribution of the location of the remote is unchanged.

Searching for the remote in the bathroom cannot reduce the entropy of the location of the remote. Another way to put it is that looking in the bathroom doesn't provide any extra information about the location of the remote, so it is the information-theoretically suboptimal decision.

This comparison would not be as cut-and-dried if the prior distribution of the location of the remote (your initial guess about where it is) was over the entire house, not just the living room. After all, there is always a small probability that the remote got misplaced outside the living room. However, the procedure of determining where to look, the portion of the house that will give you maximal information about the location of the remote, is still the same:

1. Consider about posterior distribution of the location of the remote if it is found, and compute the entropy of that distribution.
2. Compute the same entropy if the remote is not found.
3. Compute the average entropy over the two cases.
4. Repeat this computation for all locations you are considering looking in, and pick the one that gives the lowest entropy.

Entropy gives us a way to quantify our uncertainty about a quantity of interest from its probabilistic distribution in an information-theoretic manner, and this procedure leverages entropy to identify the action that maximally reduces entropy.

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